STUDENT TEAM WEEKLY REPORT: MODULE 8

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Security, Session Token Authentication, RBAC Guard & Audit Control]	**TEAM ID:** 3 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 05/10/2026 **STUDENT LEADER:** Pranav Chougule

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Cryptographic Bearer Token Engine:** Built SessionManager.java issuing unique session tokens (mmcoe_tok_<UUID>) with 2-hour rolling inactivity timeout and token validation. • **Role-Based Access Control (RBAC):** Implemented AppServer.authenticateRequest(ex, allowedRoles) returning HTTP 401 Unauthorized for missing tokens and HTTP 403 Forbidden for unauthorized roles. • **Secure Password Storage:** Enforced OpenBSD BCrypt hashing (BCrypt.gensalt(10)) in PasswordUtil.java ensuring zero plaintext passwords in PostgreSQL. • **Universal Audit Trail Logger:** Built AuditLogsHandler logging logins, question edits, quiz publishing, grading, and deletions with PostgreSQL resolution by email/username. • **Automated Security Test Suite:** Created and ran test_security_audit_module7.js verifying 401 blocking, 403 RBAC enforcement, audit log retrieval, and proctor flag persistence.	• **Unauthenticated API Access:** Administrative endpoints accessible if requested outside frontend UI. Action: Added authenticateRequest(ex, "Administrator") guarding all admin routes. **GREEN (RESOLVED)** • **Stale Session Accumulation:** Expired tokens consuming memory in long-running server instances. Action: Built periodic eviction worker cleaning expired sessions in SessionManager. **GREEN (RESOLVED)** • **Audit User ID Desynchronization:** Transient user IDs causing PostgreSQL foreign key constraint errors. Action: Added dynamic resolution querying user_id from PostgreSQL by email/username. **GREEN (RESOLVED)**	**Integration & Final Release Sign-off:** Request the Integration team to perform final regression sign-off using test_security_audit_module7.js and test_api_endpoints.py to confirm full-stack readiness.

INTEGRATION AND COLLABORATION
Database Team: Verified PostgreSQL audit_log and user tables with indexed email lookups and secure password hash constraints.
Frontend Team: Integrated sessionSecurity.js across all dashboards (admin.html, author.html, moderator.html, student.html) for RBAC redirects and Bearer header injection.
Integration Team: Executed automated security test suite test_security_audit_module7.js achieving 100% test pass rate.
Pipeline Learnings: Mastered full-stack enterprise security architectures, token lifecycle management, and regulatory audit compliance.
COMMON KPI EVIDENCES
Security Test Pass Rate: 7/7 test suites passed in test_security_audit_module7.js (HTTP 401, HTTP 403, HTTP 200, Audit Logs, and Proctor Flags).
Password Protection: 100% of stored passwords encrypted with OpenBSD BCrypt ($2a$ format, 10 rounds).
Audit Completeness: 100% of critical lifecycle events (Login, Question Edit, Quiz Publish, Grading, Delete) permanently logged with timestamps.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Pranav Chougule:** Prepare final project documentation, empirical benchmark summary, and source code archive. (Due: Oct 08, 2026) • **Pranav Chougule:** Assist team with final project presentation, demonstration recording, and submission. (Due: Oct 10, 2026)	